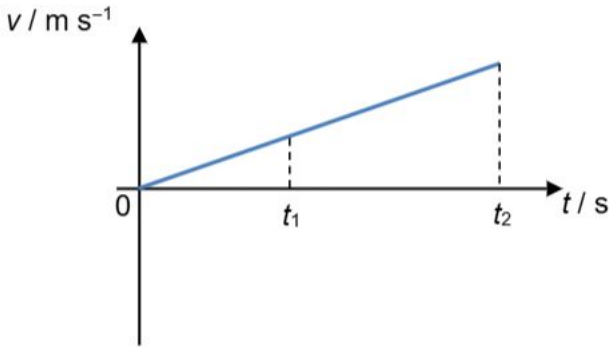
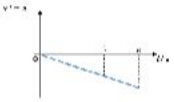
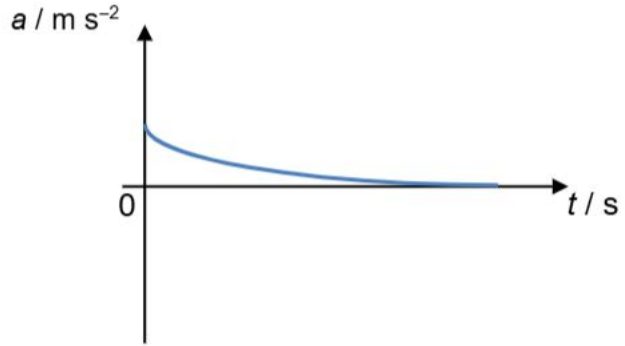
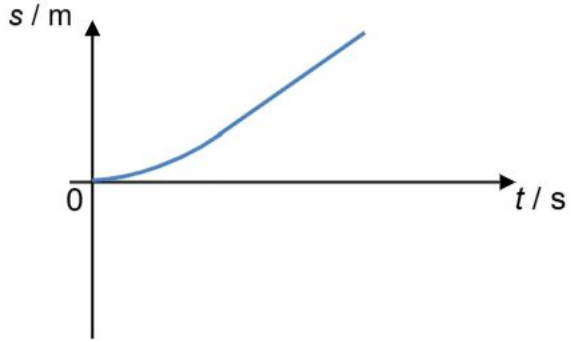
Qn	Answer	Marks
1(a)(i)	component of weight parallel to surface $= mg\sin\theta$ $ma = mg\sin\theta$ $a = g\sin\theta$ $= 9.81\sin(35°)$ $= 5.63\ \text{m s}^{-2}$	M1
1(a)(ii)	 Accept  too	A1
1(a)(iii)	area under v-t graph = distance travelled area under v-t graph $= \frac{1}{2}(u+v)t$ $= \frac{1}{2}((v-at)+v))t$ $= \frac{1}{2}(2.0-(5.63)(0.30)+2.0)(0.30)$ $= 0.3468 = 0.35\ \text{m}$ (2sf)	M1 A1
1(a)(iv)	Since power is the rate of doing work OR $P=Fv$, Constant force + increasing distance covered per unit time (or increasing speed / velocity)	B1 B1
1(b)(i)	As speed increases, the <u>air resistance</u> opposing the block's motion <u>increases</u>. When magnitude of <u>air resistance</u> is <u>equal</u> to component of block's <u>weight</u> along the slope, there is <u>no net force</u> acting on the block. By Newton's first law of motion, this results in the block travelling at constant velocity.	B1 B1

1(b)(ii)		A1
1(b)(iii)		A1
2(a)	$s_y = u_y t + \frac{1}{2} a t^2$	M1